

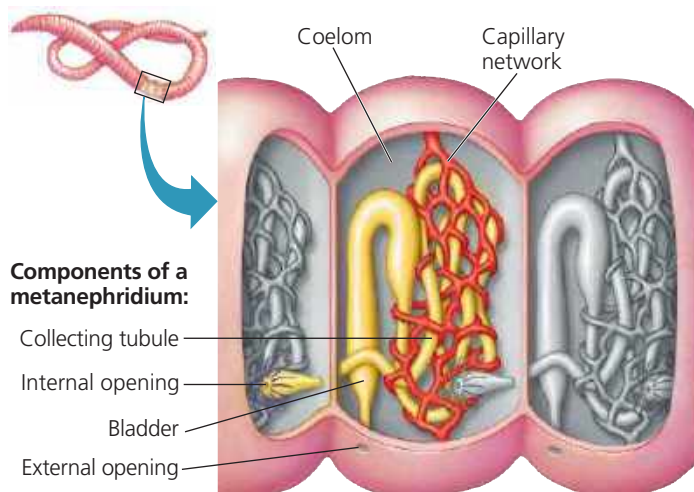
Protonephridia are also found in rotifers, some annelids, mollusc larvae, and lancelets (see Figure 34.4). In the freshwater flatworms, protonephridia serve chiefly in osmoregulation. Most metabolic wastes diffuse out of the animal across the body surface or are excreted into the gastrovascular cavity and eliminated through the mouth (see Figure 33.9). In contrast, parasitic flatworms that are isoosmotic to the surrounding fluids of their host organisms have protonephridia that primarily function in the disposal of nitrogenous wastes. Natural selection has thus adapted protonephridia to different tasks in different environments.

Metanephridia

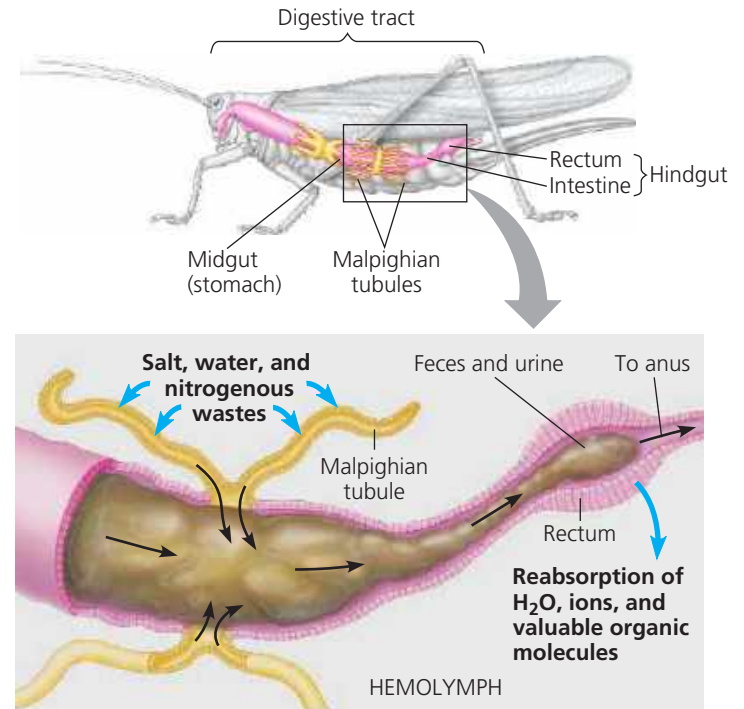
Most annelids, such as earthworms, have **metanephridia** (singular, *metanephridium*), excretory organs that collect fluid directly from the coelom (Figure 44.10). A pair of metanephridia are found in each segment of an annelid, where they are immersed in coelomic fluid and enveloped by a capillary network. A ciliated funnel surrounds the internal opening of each metanephridium. As the cilia beat, fluid is drawn into a collecting tubule, which includes a storage bladder that opens to the outside.

Earthworms inhabit damp soil and therefore usually experience a net uptake of water by osmosis through their skin. Their metanephridia balance the water influx by producing urine that is dilute (hypoosmotic to body fluids). In producing a hypoosmotic filtrate, the transport epithelium reabsorbs most solutes and returns them to the blood in the capillaries. Nitrogenous wastes, however, remain in the tubule and are excreted to the environment. The metanephridia of an earthworm thus serve both an excretory and an osmoregulatory function.

▼ **Figure 44.10 Metanephridia of an earthworm.** Each segment of the worm contains a pair of metanephridia, which collect coelomic fluid from the adjacent anterior segment. The region highlighted in yellow illustrates the organization of one metanephridium of a pair; the other would be behind it.



▼ **Figure 44.11 Malpighian tubules of insects.** Malpighian tubules are outpocketings of the digestive tract that remove nitrogenous wastes and function in osmoregulation.



Malpighian Tubules

Insects and other terrestrial arthropods have organs called **Malpighian tubules** that remove nitrogenous wastes and that also function in osmoregulation (Figure 44.11). The Malpighian tubules extend from dead-end tips immersed in hemolymph to openings into the digestive tract. The filtration step common to other excretory systems is absent. Instead, the transport epithelium that lines the tubules secretes certain solutes, including nitrogenous wastes, from the hemolymph into the lumen of the tubule. Water follows the solutes into the tubule by osmosis.

As fluid passes from the tubules into the rectum, most solutes are pumped back into the hemolymph; water reabsorption by osmosis follows. The nitrogenous wastes—mainly insoluble uric acid—are eliminated as nearly dry matter along with the feces. The insect excretory system is capable of conserving water very effectively, a key adaptation contributing to the tremendous success of insects on land.

Kidneys

In vertebrates and some other chordates, a compact organ called the **kidney** functions in both osmoregulation and excretion. Like the excretory organs of most animal phyla, kidneys consist of tubules. The tubules of kidneys are